Srividya Chekuri

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SUMMARY:

- Results-driven Data Engineer with 3+ years of experience designing and implementing scalable data solutions across healthcare and banking industries.
- Designed and implemented ETL pipelines, automating data processing workflows for improved efficiency.
- Utilized Power BI, Excel, and Denodo to develop interactive reports and dashboards for business insights.
- Developed and optimized machine learning workflows, enhancing predictive analytics capabilities.
- Leveraged cloud platforms like Azure and AWS to build and manage large-scale data architectures.
- Adept at communicating effectively with stakeholders, aligning technical solutions with strategic business goals.
- Strong expertise in data pipeline development, ETL automation, and machine learning workflows, leading to optimized business decision-making.
- Experienced at leveraging predictive analytics to enhance operational efficiency, reduce costs, and improve data accessibility for analytics.

TECHNICAL SKILLS:

Programming Languages: Python, R, Spark, Java

Databases: MySQL, SQL Server, MongoDB, PostgreSQL, SQLite, Neo4j

Data Analysis: Data Cleaning, Feature Engineering, Dimensionality Reduction, Time Series Analysis

Data Visualization: Power BI, Tableau

Machine Learning & AI: Regression, Decision Trees, Clustering, Deep Learning (CNN, RNN, LSTM, GANs, Transformers), NLP, Generative AI (RAG, Agentic Functions, LLMs)

Cloud & Big Data: Azure (Data Factory, Databricks, Synapse, ML, Fabric, AKS), AWS (EC2, Redshift, Sage Maker, S3, EKS), GCP (Big Query, Dataflow)

Scripting & Automation: Shell, Bash, Terraform, CI/CD (Jenkins, GitHub Actions)

Version Control: Git, DVC, SVN

Development Methodologies: Agile, SDLC

PROFESSIONAL EXPERIENCE:

Data Engineer, FedEx, August 2023 - Present

Project: Designed, developed, and optimized a cloud-based inventory management system by leveraging advanced data engineering and machine learning techniques. The solution streamlined inventory tracking, enhanced demand forecasting accuracy, and minimized stockout risks, ultimately leading to improved operational efficiency and cost savings.

- Developed and maintained ETL pipelines using Azure Data Factory to extract, transform, and load data from multiple sources into Azure Synapse Analytics and Snowflake, ensuring efficient data integration and storage.
- Implemented data cleansing and preprocessing techniques, achieving over 95% data accuracy by reducing missing values by 80% and outliers by 90%, thereby improving the reliability of predictive models.
- Designed and deployed machine learning-driven forecasting models, incorporating deep learning techniques and SARIMA models to enhance demand predictions, improving forecast accuracy by 20%.
- Developed real-time monitoring dashboards in Power BI, providing stakeholders with comprehensive visibility into key inventory metrics, including turnover rates, stockouts, and replenishment trends.
- Automated ML model deployment and management using Jenkins CI/CD pipelines, ensuring continuous integration and deployment, reducing manual intervention, and improving model efficiency by 30%.
- Integrated Generative AI and NLP models, including Llama 2 and GPT, to analyze supplier feedback and predict potential supply chain disruptions, allowing for proactive adjustments to procurement strategies.
- Utilized Neo4j for vector-based search and graph analytics to analyze relationships between products, demand fluctuations, and supplier performance, enhancing data retrieval and insight generation.
- Conducted root cause analysis for data discrepancies, implementing automated data validation rules to ensure high-quality, consistent data ingestion into machine learning workflows.
- Partnered with cross-functional teams, including data scientists, business analysts, and supply chain managers, to refine machine learning models, aligning insights with operational goals.
- Achieved a 30% reduction in excess inventory levels and a 25% decrease in carrying costs, optimizing warehouse operations through data-driven inventory management strategies.

Data Engineer, Elico Health Care Services, India, India, October 2020 to May 2022

Project: Developed and implemented a robust data engineering framework that facilitated real-time analytics and predictive modeling for fraud detection and customer segmentation within digital banking and healthcare. This solution enhanced risk assessment strategies, improved customer retention models, and ensured compliance with stringent data privacy regulations.

- Processed and analyzed 5M+ customer transactions and healthcare claims using SQL and Python, ensuring accurate and efficient data extraction, transformation, and storage.
- Designed and implemented feature engineering strategies, enhancing the predictive power of machine learning models and improving overall data pipeline efficiency by 40%.
- Created and optimized interactive dashboards in Tableau and Power BI, providing real-time insights into fraud detection patterns, customer behavior, and business performance metrics.
- Developed automated data validation processes, reducing data processing errors by 50% and enhancing the reliability of analytical reports.
- Improved customer segmentation models by 35%, leveraging machine learning to refine marketing strategies and personalized engagement initiatives.
- Ensured compliance with HIPAA and GDPR regulations, implementing robust security and privacy measures to protect sensitive healthcare and financial data.
- Built fraud detection algorithms using AWS SageMaker, improving fraud detection accuracy by 25% and mitigating financial risks associated with fraudulent activities.
- Designed scalable ETL workflows, efficiently processing structured and unstructured datasets, optimizing query performance, and supporting large-scale analytics applications.
- Delivered customized analytical reports to stakeholders, synthesizing insights from various data sources to drive strategic decision-making and operational improvements.
- Collaborated with business analysts, data scientists, and engineering teams to refine data-driven solutions, enhancing fraud detection mechanisms and customer insights for business optimization

ACADEMIC PROJECTS:

- **Current Real-World Job Skill Demand vs. Near-Future Graduate Supply:** Analysed industry trends and workforce demand using Python, Power BI, and Excel, identifying key skill gaps to align academic curricula with future job market needs.

- **Vehicle Number Plate Detection:** Developed an OCR-based machine learning model using OpenCV and Python, achieving 90%+ accuracy in real-time number plate detection and recognition.

- **Automated Resume Screener:** Created an AI-powered ATS using NLP and machine learning, reducing recruitment processing time by 50% and improving candidate-job matching efficiency.

EDUCATION:

M.S Data Science - University of New Haven, Dec 2023

Bachelor Of Technology, Electronics & Comm. Engineering **-** Chalapathi Institutes of Engineering & Technology, Apr 2021